

What's Happening in Geological T&M

Today, test and measurement is moving more towards software-defined approach. Geological T&M is no exception as it enables scientists to make the best use of the time and effort that they put in taking these measurements. This approach enables engineers to get more flexibility and greater performance at lower cost

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Geological testing or geotechnical investigations are performed by geotechnical engineers or engineering geologists to unearth information about the physical properties of soil and rock around a site. The information is crucial when trying to design earthworks and foundations for the proposed structures and for repair of distress to earthworks and structures caused by surface conditions. Geotechnical investigations are also used to measure other parameters

like the thermal resistivity of soil, back-fill materials required for underground transmission lines, oil and gas pipelines, radioactive waste disposal and solar thermal storage facilities.

A geotechnical investigation is a process in which the physical properties of a site are assessed for the purpose of determining what kind of work can be done on the site safely. Before land can be developed or redeveloped, a geotechnical investigation of the concerned site is recommended as a safety procedure. This process is also required or recommended in the

wake of incidents like earthquakes, the emergence of foundation cracks on land that was thought to be solid and so forth. The goal of such an investigation is to confirm that the land is safe to work on.

Sometimes geophysical methods are used to obtain data about sites. Sub-surface exploration usually involves soil sampling and laboratory tests of the soil samples retrieved. This can include complex processes of geological mapping, geophysical methods and photogrammetry,

or it can be as simple as a walking around the site to observe its physical conditions.

Need for geotechnical investigation

Geotechnical investigation is gaining prominence for a number of reasons. Main among them is the increased activity in construction—not just in terms of volume but also complexity. Gone are the days when soil was a decisive factor, determining the kind of construction to be done at a place. Nowadays, the soil is made suitable for the project.

Geotechnical services encompass different functions, each as varied as the next, all of which involve measuring the effect of earth and soil on construction projects. Geotechnical laboratories may analyse soil samples at a proposed construction site to help assess whether the ground can support the needed amount of weight. Other types of inspections may also be conducted on a site to estimate the likelihood of potential damage from natural disasters such as earthquakes and landslides.

Majority of engineering construction projects begin with a thorough analysis of the proposed building site, which includes soil assessment. Geotechnical service professionals often run laboratories that conduct detailed analysis of the earth on location to provide a list of all the soil components and their proportions. The research also contains information on other soil characteristics such as whether it is rocky, holds a lot of moisture or is very compacted. Engineers use this data to determine earth parameters such as its weight capacity, permeability and

